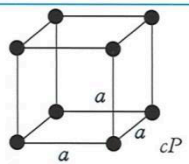
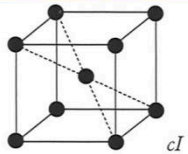
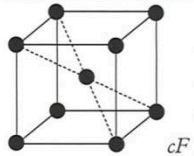
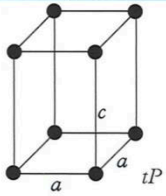
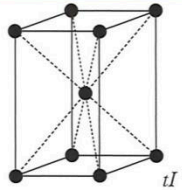
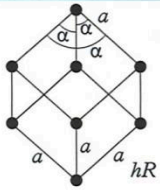
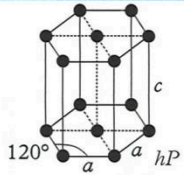
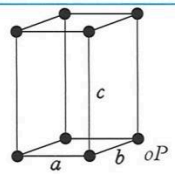
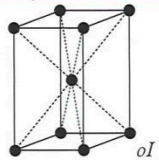
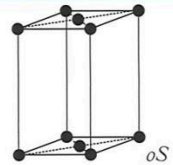
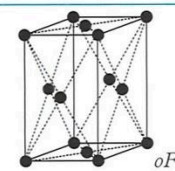
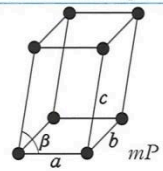
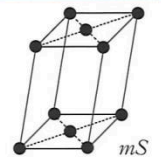
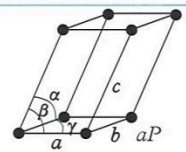
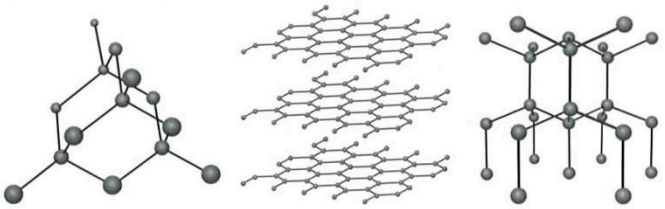
kristalrooster → / *eenheidscel ↓*		*primitief (eenvoudig)* *P*	*lichaamsgecentreerd* *I*	*grondvlakgecentreerd* *S*	*vlakgecentreerd* *F*
kubisch = isometrisch	*c*				
abundantie van deze structuur in kristallen:					
12%		cesiumjodide CsI	ferriet Fe		aluminium Al
tetragonaal	*t*				
10%		wulfeniet $PbMoO_4$	martensiet Fe,C		
romboëdrisch, trigonaal	*h*				
9%		calciet $CaCO_3$			
hexagonaal	*h*				
8%		beril $Be_3Al_2Si_6O_{18}$			

vervolg

kristalrooster → / *eenheidscel ↓*		*primitief (eenvoudig)* *P*	*lichaamsgecentreerd* *I*	*grondvlakgecentreerd* *S*	*vlakgecentreerd* *F*
orthorombisch	*o*				
22%		andalusiet Al_2SiO_5	dawsoniet $NaAlCO_3(OH)_2$	ferruciet $NaBF_4$	melanothalliet Cu_2OCl_2
monoklien	*m*				
32%		zwavel S		glauber $Na_2SO_4 \cdot 10H_2O$	
triklien	*a*				
7%		wollastoniet $CaSiO_3$			

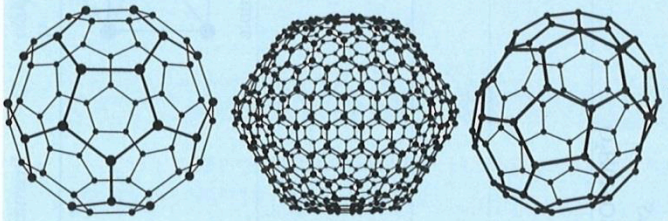
netwerkstructuren



a b c

a diamant (kubisch)
b grafiet (gelaagd)
c lonsdaleïet (hexagonaal)
slijpen sieraden, sleepcontacten, moderator in kernreactor, meteoriet

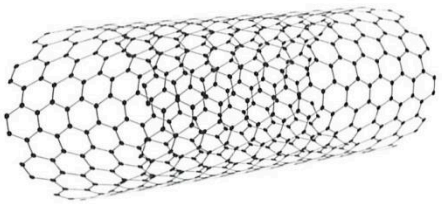
moleculaire (nano) structuren: fullerenen



d e f

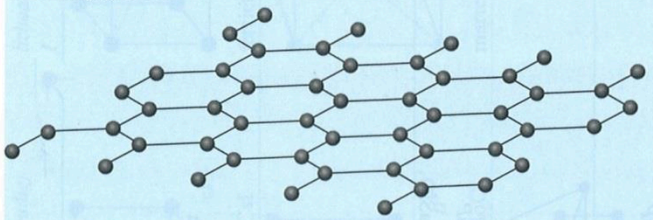
d C_{60}
e C_{540}
f C_{70}
farmacie, opsluiten atomen, supergeleiding

macromoleculaire structuur: nanobuisjes



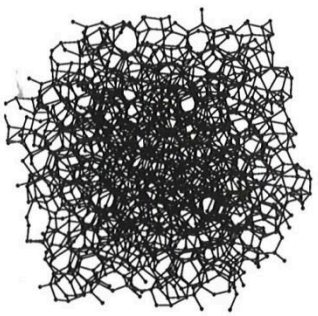
variabele lengte en diameter
AF-microscopie, sensoren, waterstofopslag, zonnecel, supersterke materialen, nanolager, bactericide

monolaag: grafeen

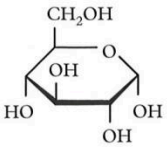


quantumeigenschappen: *DNA-analyse, computerchips*

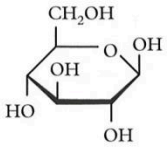
amorf



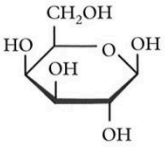
houtskool, steenkool
cokes, roet
'nieuwe koolstof' die in elke richting enorme druk kan doorstaan
technische aambeelden (i.p.v. technisch diamant)



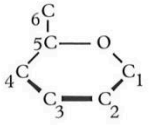
D-glucose
(α-cycloformule)
(druivensuiker)



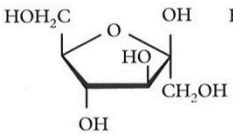
D-glucose
(β-cycloformule)
(druivensuiker)



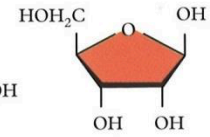
D-galactose
(β-cycloformule)



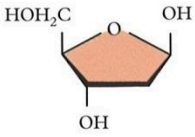
nummering
C-atomen



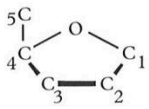
D-fructose
(β-cycloformule)
(vruchtensuiker)



D-ribose[1]
(β-cycloformule)



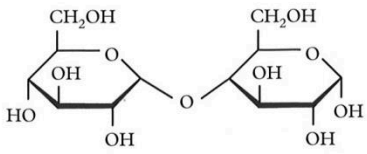
D-2-deoxyribose[1]
(β-cycloformule)



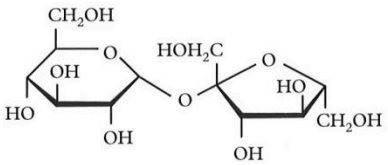
nummering
C-atomen
ribose en deoxyribose

1 ▶ Betekenis kleur: zie tabel 71C.

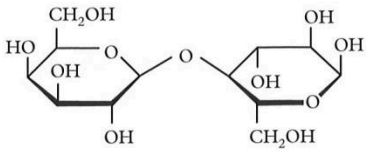
Disachariden 2



maltose (moutsuiker)



sacharose (bietsuiker, rietsuiker)

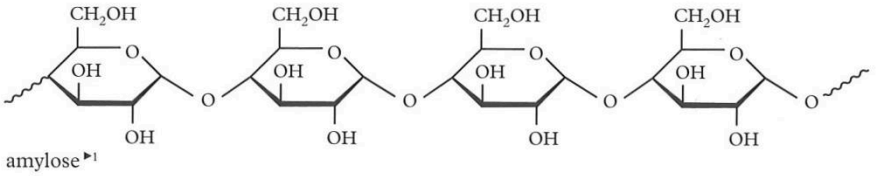


lactose (melksuiker)

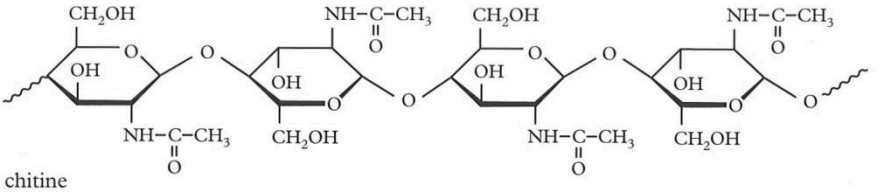
■ Mono- en disachariden worden vaak aangeduid als suikers.

F Sachariden, koolhydraten

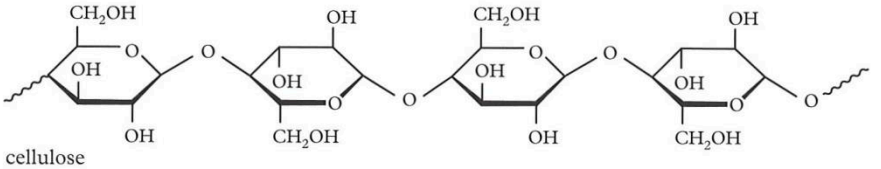
3 Polysachariden



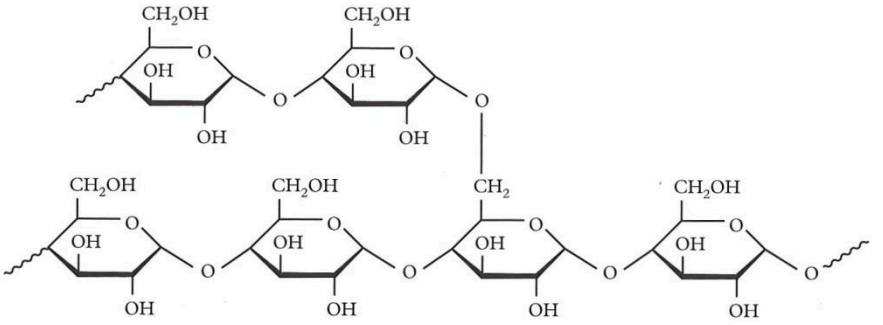
amylose ▶1



chitine



cellulose

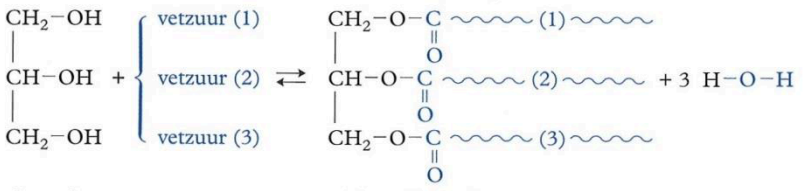


glycogeen: sterk vertakte keten
amylopectine: zwak vertakte keten

■ Veel sachariden voldoen aan de algemene formule $C_m(H_2O)_n$; de sachariden worden daarom vaak aangeduid als koolhydraten.
1 ▶ Zetmeel bestaat uit amylose en amylopectine.

G Vetten, vetzuren en fosfolipiden

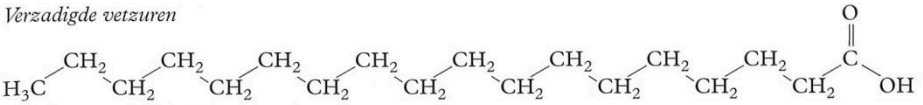
1 Vetten

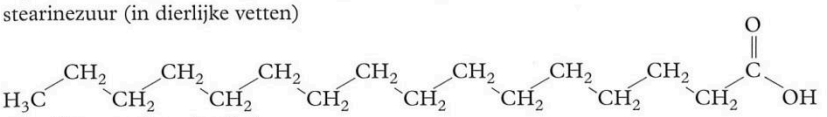


glycerol — triglyceride (vet)

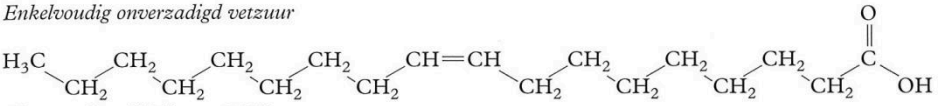
Vetzuren 2

Verzadigde vetzuren

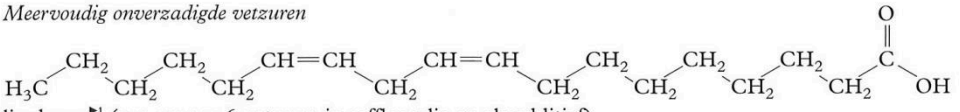

stearinezuur (in dierlijke vetten)

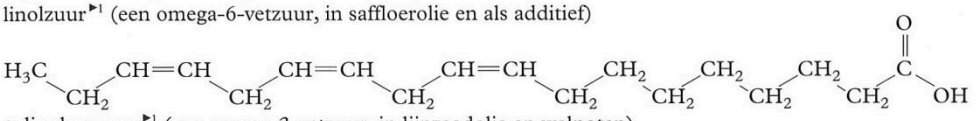

palmitinezuur (in palmolie)

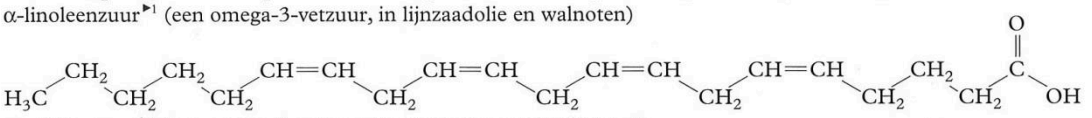
Enkelvoudig onverzadigd vetzuur


oliezuur (in olijfolie tot 80%)

Meervoudig onverzadigde vetzuren

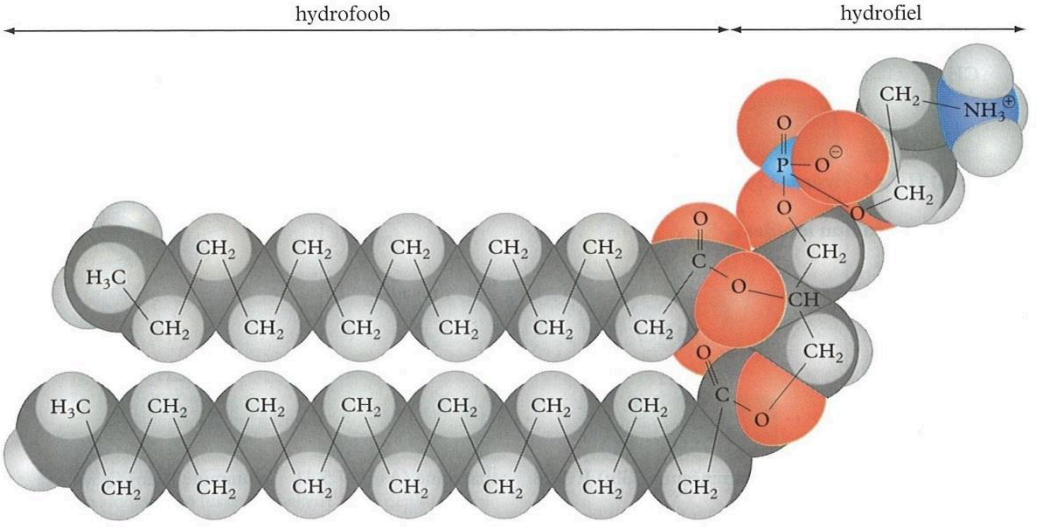

linolzuur[1] (een omega-6-vetzuur, in saffloerolie en als additief)

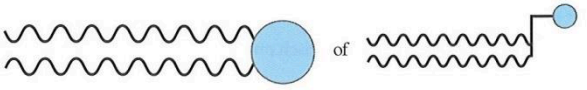

α-linoleenzuur[1] (een omega-3-vetzuur, in lijnzaadolie en walnoten)


arachidonzuur[1] (een omega-6-vetzuur, in eierdooiers en wild(vlees))

1 ▶ essentieel vetzuur

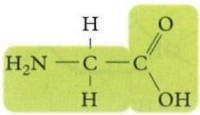
Fosfolipiden 3

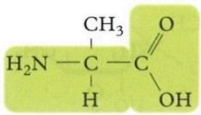


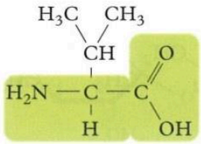
■ symbolische notatie van fosfolipide  of

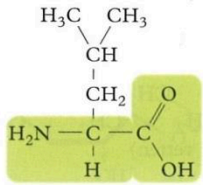
Aminozuren van eiwitten[1]

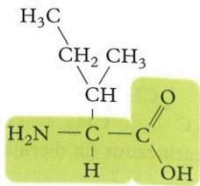
alifatische aminozuren


glycine (Gly) G 6,06 [1]

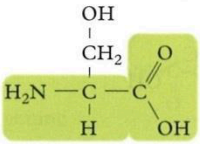

alanine (Ala) A 6,11

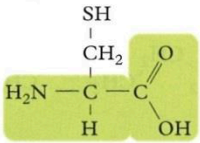

valine (Val) V 6,00 [2]

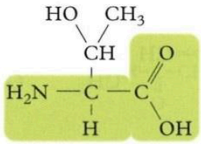

leucine (Leu) L 6,04 [2]

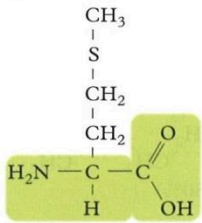

isoleucine (Ile) I 6,04 [2]

aminozuren met hydroxyl- of zwavelhoudende zijketen

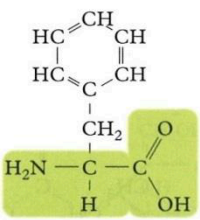

serine (Ser) S 5,68

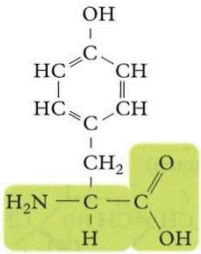

cysteïne (Cys) C 5,02

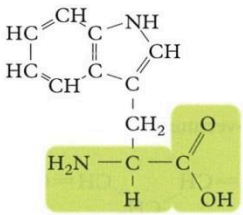

threonine (Thr) T 5,60 [2]


methionine (Met) M 5,74 [2]

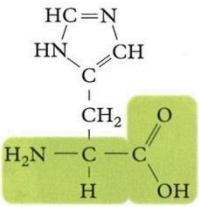
aromatische aminozuren

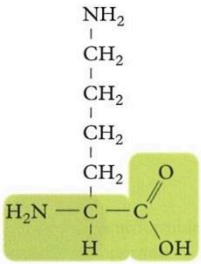

fenylalanine (Phe) F 5,91 [2]

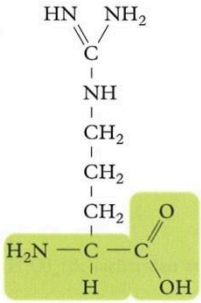

tyrosine (Tyr) Y 5,63


tryptofaan (Trp) W 5,88 [2]

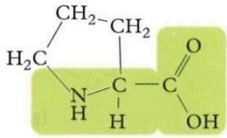
basische aminozuren


histidine (His) H 7,64 [2]

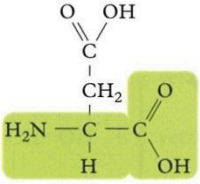

lysine (Lys) K 9,47 [2]

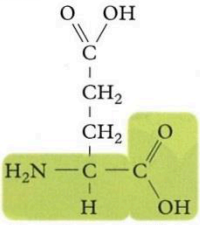

arginine (Arg) R 10,76

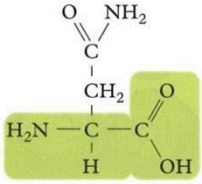
cyclisch aminozuur

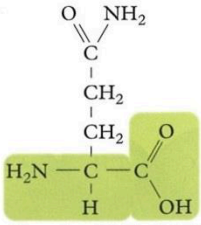

proline (Pro) P 6,30

zure aminozuren en hun amides


asparaginezuur (Asp) D 2,98

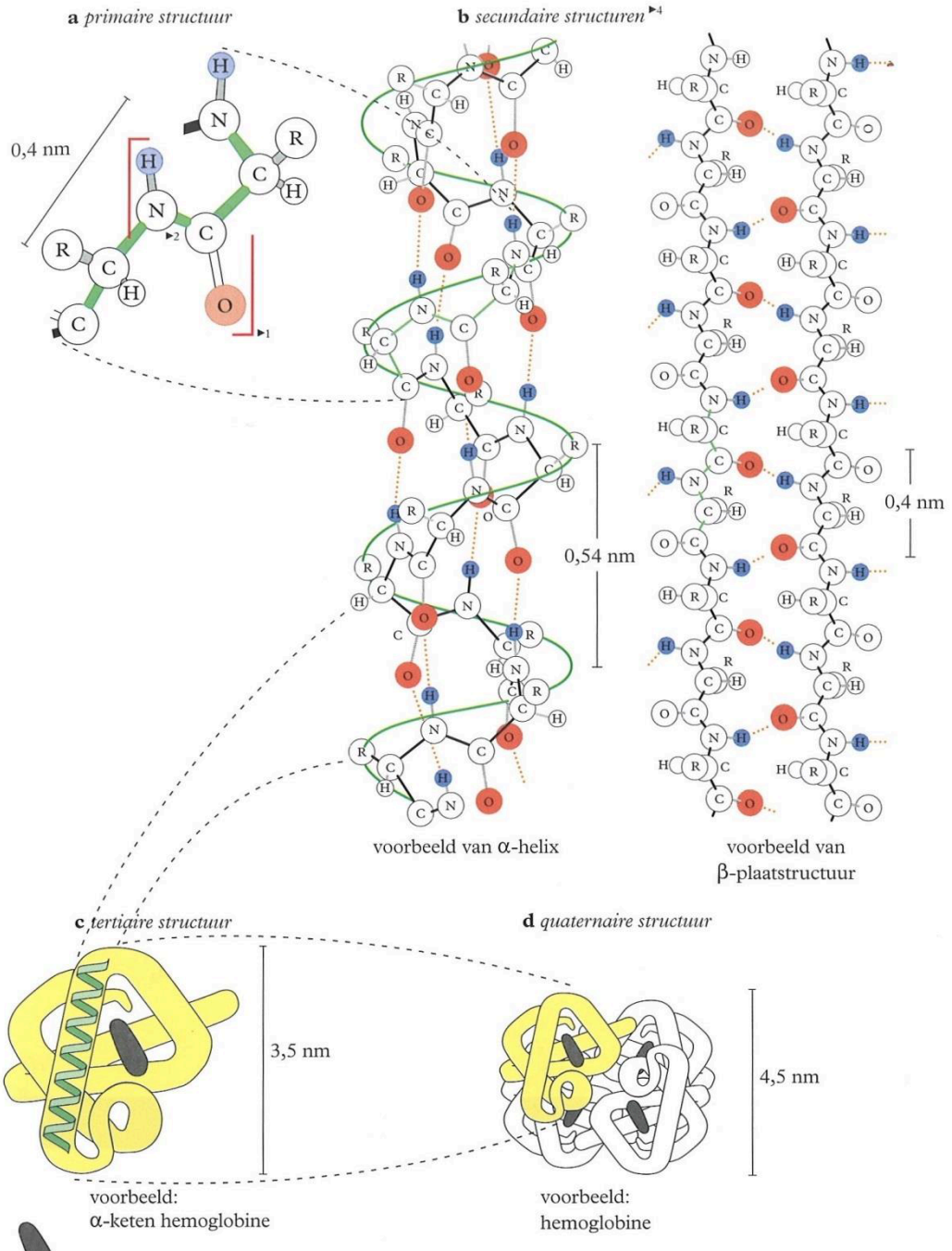

glutaminezuur (Glu) E 3,08


asparagine (Asn) N 5,41


glutamine (Gln) Q 5,65

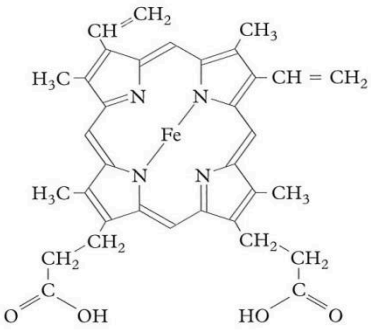
■ Bijbehorende codons in het mRNA; zie tabel 71G.

1 ▶ naam, (3-lettersymbool), 1-lettersymbool, isoëlektrisch punt pI
2 ▶ voor mensen essentieel

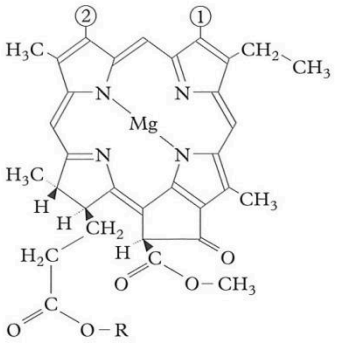


1 ▶ De atoomgroep tussen de haken [ ] is de peptidegroep.
2 ▶ peptidebinding
3 ▶ Heem: zie tabel 67I.
4 ▶ Plaatstructuur kan uit meer ketens bestaan.

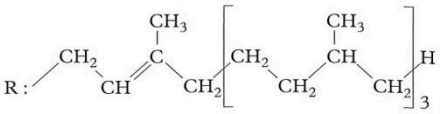
I Heem, fotopigmenten



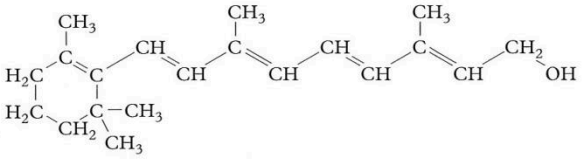
heem



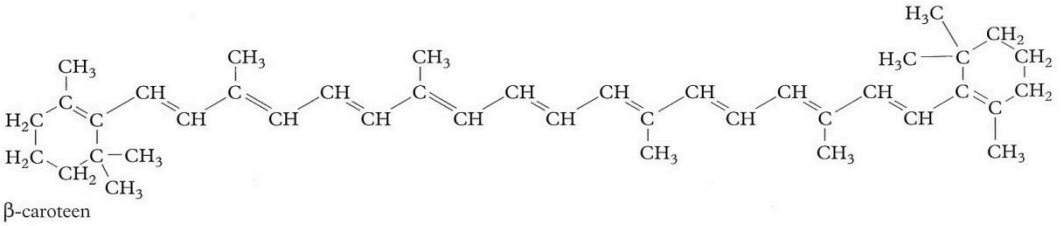
chlorofyl



①	$-CH_3$	bij chlorofyl-a en -d
	$-CHO$	bij chlorofyl-b
②	$-CH=CH_2$	bij chlorofyl-a en -b
	$-CHO$	bij chlorofyl-d

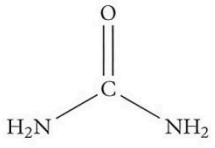


vitamine A

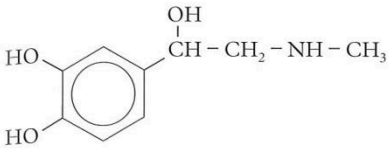


β-caroteen

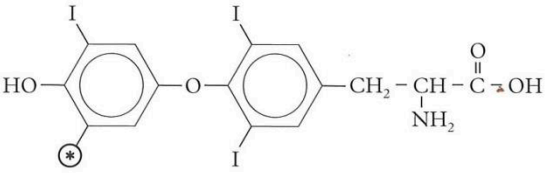
J Ureum



Derivaten van tyrosine 1

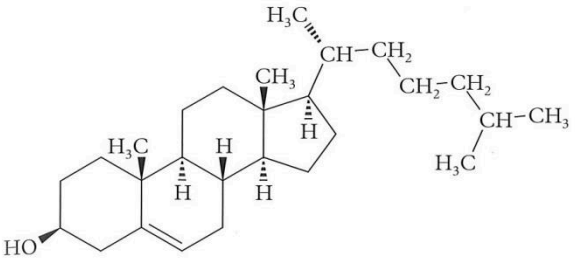


adrenaline, epinefrine

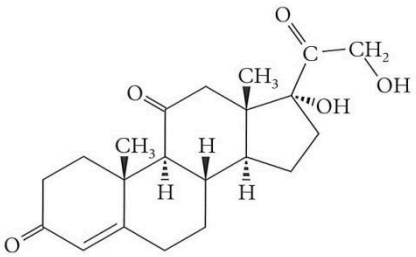


—⊛ = –I: thyroxine
—⊛ = –H: trijoodthyronine

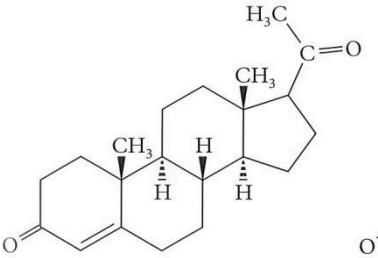
Steroïden 2



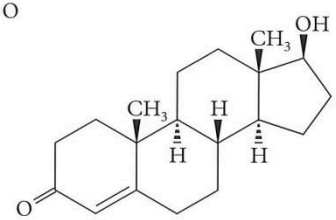
cholesterol ▸1



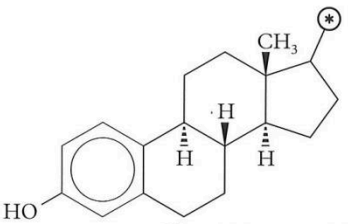
cortison ▸2



progesteron ▸2



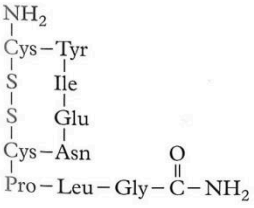
testosteron ▸2



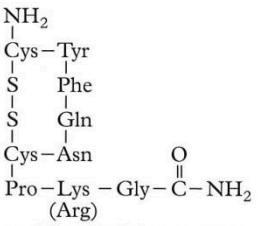
oestradiol

—⊛ = –OH: oestradiol ▸2
—⊛ = =O: oestron ▸2

Peptidehormonen 3



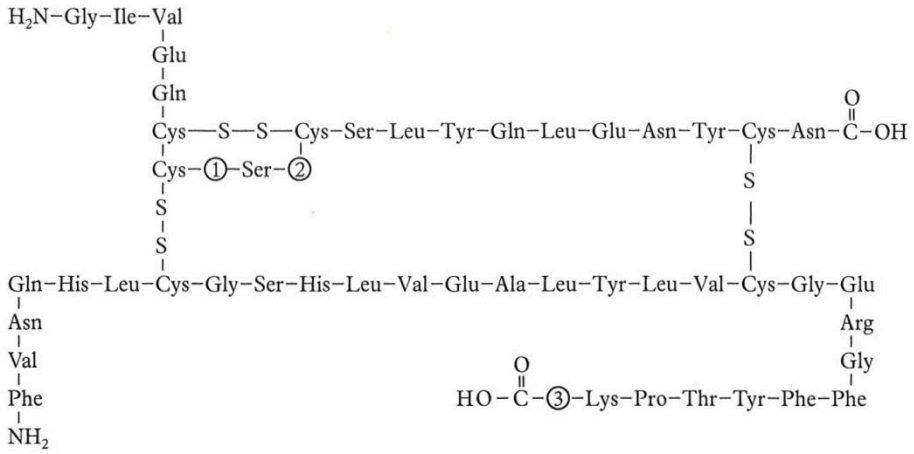
ocytocine (oxytocine)



antidiuretisch hormoon (vasopressine)

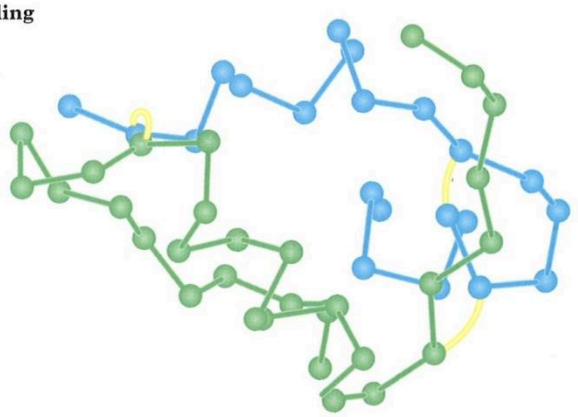
■ Voor werking van de hormonen: zie tabel 89A.

1 ▸ uitgangsstof voor steroïdhormonen
2 ▸ steroïdhormonen

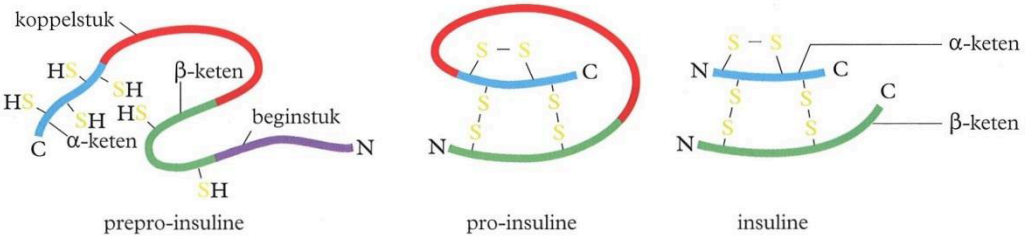


- menseninsuline: 1 = threonine 2 = isoleucine 3 = threonine
- varkensinsuline: 1 = threonine 2 = isoleucine 3 = alanine
- runderinsuline: 1 = alanine 2 = valine 3 = alanine

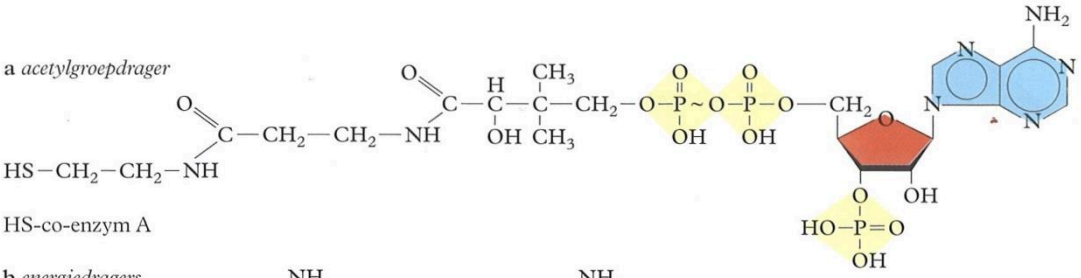
ruimtelijke voorstelling



wordingsgeschiedenis

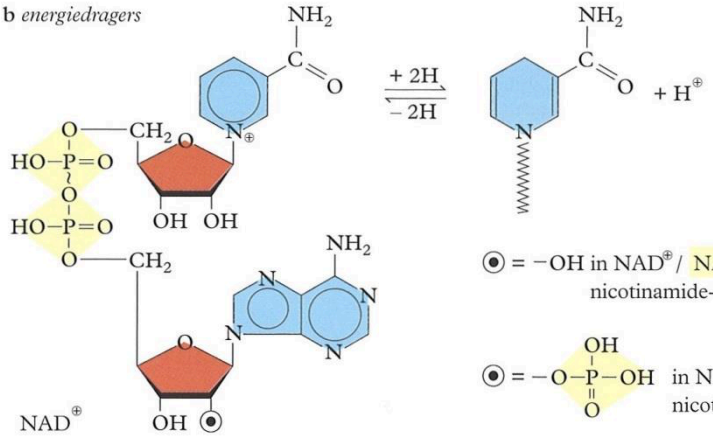


a *acetylgroepdrager*



HS-co-enzym A

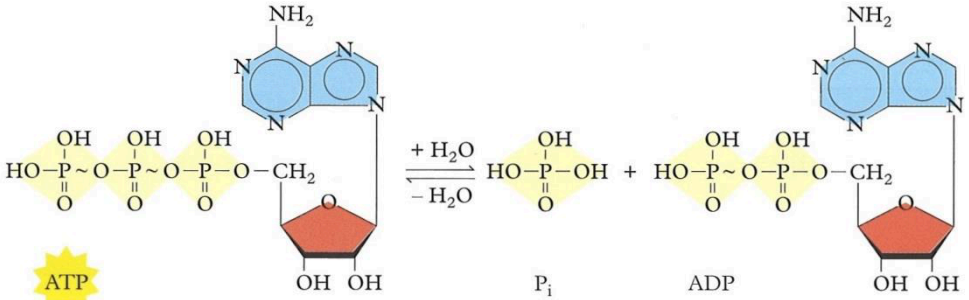
b *energiedragers*



NAD$^{\oplus}$

⦿ = −OH in NAD$^{\oplus}$ / NADH,H$^{\oplus}$
nicotinamide-adenine-dinucleotide

⦿ = in NADP$^{\oplus}$/ NADPH,H$^{\oplus}$
nicotinamide-adenine-dinucleotide-fosfaat



ATP
adenosinetrifosfaat

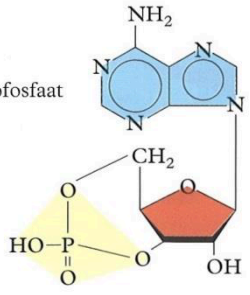
P_i

ADP
adenosinedifosfaat

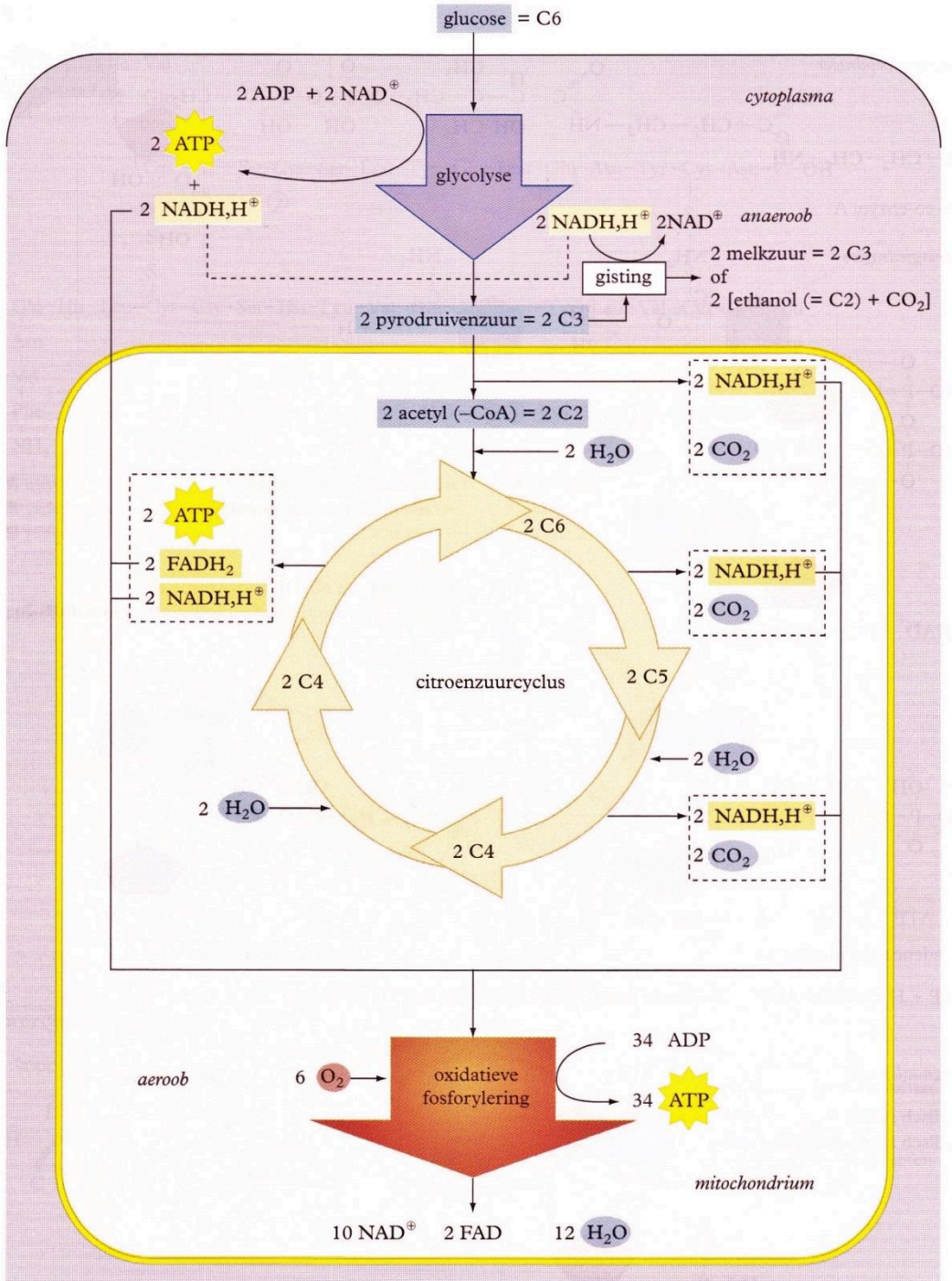
$ATP + H_2O \rightleftarrows P_i + ADP$ maximale energieopbrengst in een cel: $31 \cdot 10^3$ J mol^{-1}

c *signaaldrager*

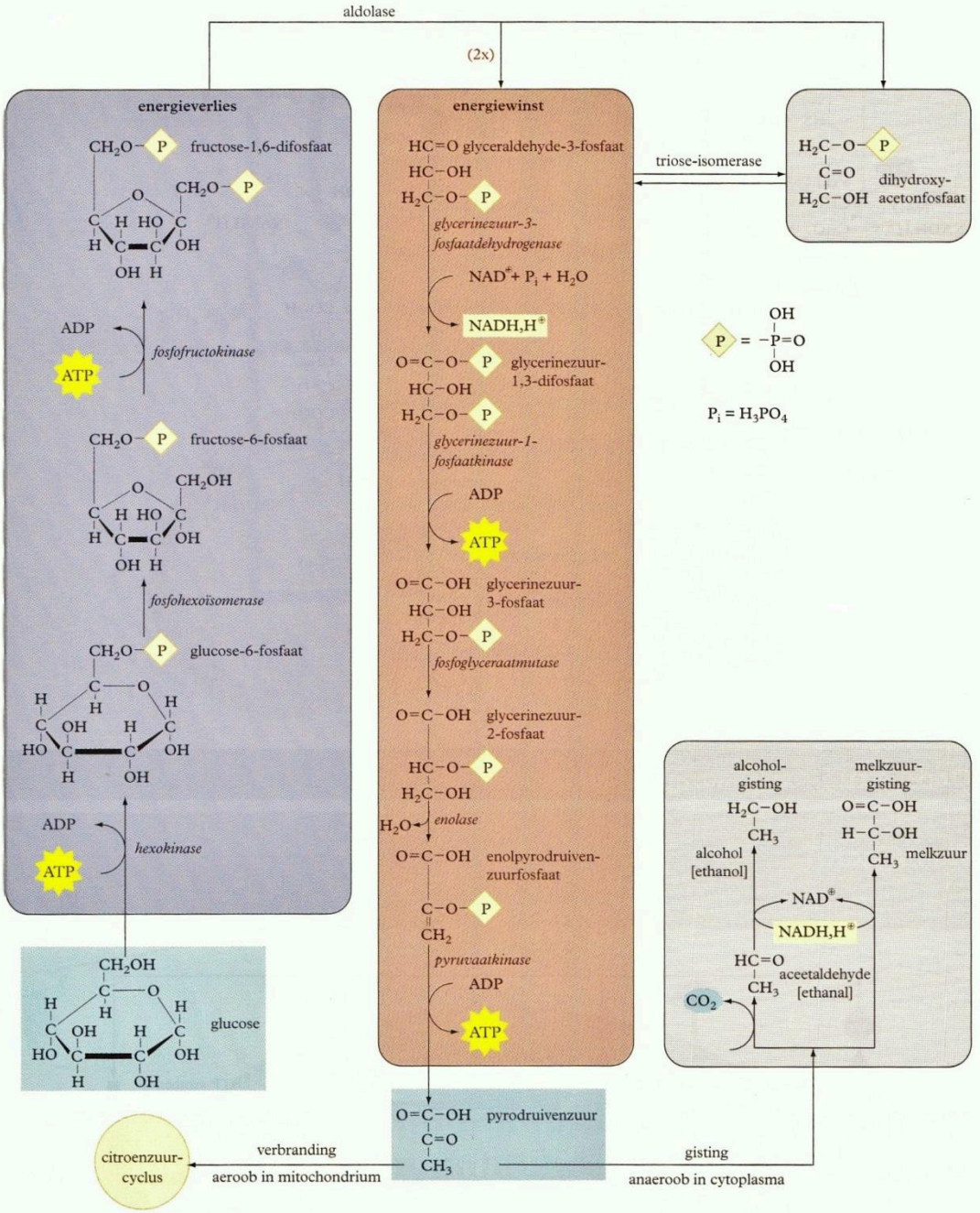
cyclisch AMP
cyclisch adenosinemonofosfaat



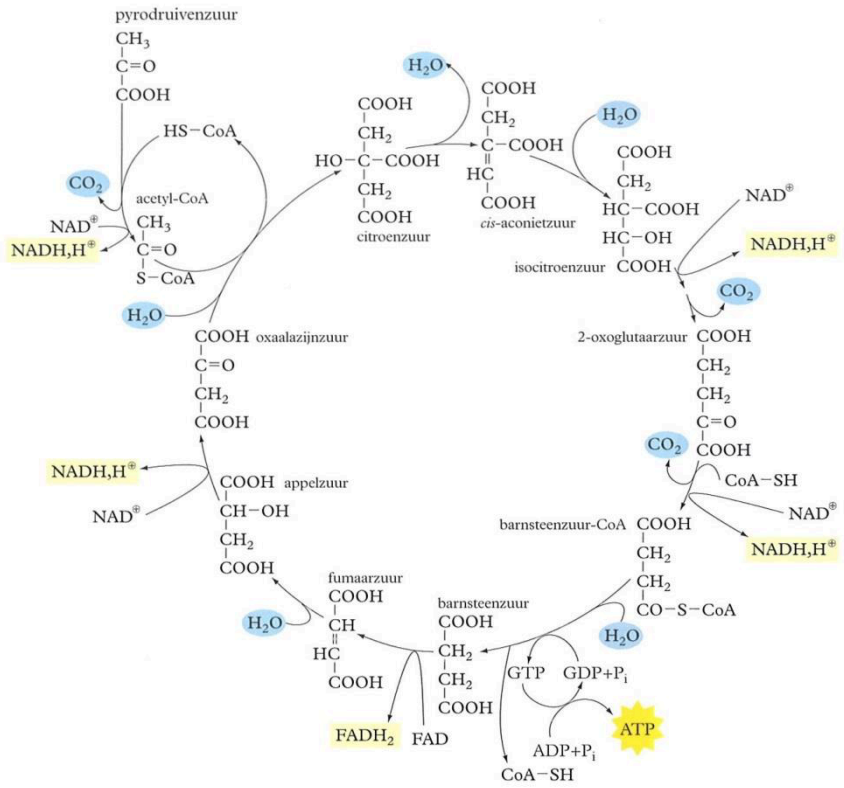
- ~ = symbool voor een energierijke binding
- P_i = anorganisch fosfaat



■ Bij het in het cytoplasma geproduceerde NADH,H⊕ wordt bij het passeren van de mitochondriummembraan 1 ATP gebruikt.


aldolase
(2x)
energieverlies
energiewinst
fructose-1,6-difosfaat
ADP
ATP
fosfofructokinase
fructose-6-fosfaat
fosfohexoïsomerase
glucose-6-fosfaat
hexokinase
glucose
glyceraldehyde-3-fosfaat
triose-isomerase
dihydroxy-acetonfosfaat
glycerinezuur-3-fosfaatdehydrogenase
NAD+ + Pi + H2O
NADH,H+
glycerinezuur-1,3-difosfaat
glycerinezuur-1-fosfaatkinase
glycerinezuur-3-fosfaat
fosfoglyceraatmutase
glycerinezuur-2-fosfaat
enolase
enolpyrodruivenzuurfosfaat
pyruvaatkinase
pyrodruivenzuur
P = −P=O
Pi = H3PO4
alcoholgisting
melkzuurgisting
alcohol [ethanol]
melkzuur
aceetaldehyde [ethanal]
CO2
citroenzuurcyclus
verbranding
aeroob in mitochondrium
gisting
anaeroob in cytoplasma

C Citroenzuurcyclus



D Oxidatieve fosforylering, ademhalingsketen

